Do Discourse Markers Signal Speaker Gender? Corpus Statistics and BERT Classifiers on the Spoken BNC2014

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Abstract

Popular and early sociolinguistic accounts claim that women hedge more and use more discourse markers (DMs) than men. We test how much speaker gender is recoverable from DM use in the Spoken BNC2014, a corpus of 11.5 million words of informal British conversation. Treating speakers rather than words as the unit of analysis, only 10 of 53 DMs differ reliably between women and men, compared with 37 under pooled frequency tests. Women use more boosters (*really*, *so*) and interactional markers (*oh*), men use more of the filled pause *er*, and hedging does not differ. A speaker's overall DM profile predicts their gender with 0.726 balanced accuracy, but single utterances barely do: on unseen speakers, fine-tuned BERT reaches 0.576 accuracy on full utterances and 0.541 on DMs alone, and removing DMs does not reliably lower accuracy. DM use is gendered on average, but far too weakly to identify an individual from what they say in a single turn.

1 Introduction

Discourse markers such as *like*, *you know*, *I mean* and *well* organize talk, signal stance and manage the relationship between speakers (Schiffrin, 1987). Because they are frequent, easy to identify and socially meaningful, they have long been a site for claims about gendered speech. Lakoff (1973) described women's language as tentative, hedged and full of intensifiers, and that characterization still circulates widely. Later work showed that the picture is more complicated: the same form can serve different functions for different speakers (Holmes, 1986), many reported differences are small or context-dependent (Cameron, 2007), and corpus-wide generalizations can be driven by a handful of individuals (Brezina and Meyerhoff, 2014).

Pre-trained language models offer a new way to ask the question. If gender were strongly encoded in DM use, a model like BERT (Devlin et al., 2019) fine-tuned to predict speaker gender should be able to recover it from DMs alone. Classifier accuracy is also easy to over-interpret: a model trained on whole utterances may succeed because of topic words rather than DMs, and a model tested on utterances from speakers it saw in training may be recognizing individuals rather than gender.

This paper is designed around those pitfalls. Our contributions are:

1. Frequency comparisons for 53 DMs in five functional categories at both the corpus and the speaker level, showing that most pooled-count differences do not survive a speaker-level test.

2. A speaker-level classifier that predicts gender from a speaker's DM rates, tested against a permutation baseline.

3. Utterance-level classifiers evaluated on unseen speakers, with ablations that isolate DMs (*full text*, *DMs removed*, *DMs only*) and speaker-clustered confidence intervals.

4. A masked-language-model probe showing that the gender associations pre-trained BERT holds for DMs correlate with real usage.

Code and all generated outputs are available at https://github.com/iahmed20/DMs-w-BERT.

2 Related Work

Gender and discourse markers. Lakoff (1973) argued, largely from introspection and personal observation, that women use more hedges and intensifiers and that this weakens their perceived authority. Holmes (1986) showed that the function of *you know* differs by gender more than its frequency does. Corpus studies have since found reliable but modest differences for specific forms. In British

English, Tottie (2011) found that *uh* and *um* pattern differently by gender, age and social class, and Laserna et al. (2014) found that filled pauses and DMs such as *like*, *I mean* and *you know* vary with gender, age and personality. Cameron (2007) reviews the broader evidence and argues that variation within each gender dwarfs the differences between them.

Statistics for sociolinguistic corpora. Pooled frequency tests such as log-likelihood (Rayson and Garside, 2000) treat every token as independent. Brezina and Meyerhoff (2014) show that this assumption is badly violated in speaker-annotated corpora, where a few prolific speakers can create apparently significant differences, and recommend tests on per-speaker rates. We report both.

Predicting gender from text. Bamman et al. (2014) predicted author gender from social-media vocabulary with high accuracy, but found that the most predictive features were topical and that many individuals do not follow their gender's aggregate pattern. Our DMs-only and DMs-removed conditions separate DM signal from this kind of topical signal.

Gender in pre-trained language models. Template-based masked-language-model probes have been used to measure gender associations in BERT (Kurita et al., 2019), and several studies document and try to remove such associations (Bhardwaj et al., 2021; Li et al., 2021). Sileo et al. (2020) analyze the semantics of DMs with neural models. We use a template probe to ask whether the associations BERT learned in pre-training line up with gender differences in real conversation.

3 Data

3.1 The Spoken BNC2014

The Spoken BNC2014 (Love et al., 2017) contains 11.5 million words of informal conversation recorded by members of the public in the United Kingdom, transcribed and annotated with CLAWS7 part-of-speech tags. Speaker metadata, including gender, is self-reported. We use the tagged XML release and treat each `<u>` element as one utterance.

We keep utterances from speakers whose metadata records gender as female or male: 365 women contributing 7,074,706 tokens and 305 men contributing 4,348,983 tokens. Utterances attributed to unidentified speakers (`UNKFEMALE`, `UNKMALE`, `UNKMULTI`) are excluded, because they cannot be assigned to an individual for speaker-level analysis or speaker-disjoint splitting. All tokens are lowercased. Transcription and anonymization codes (e.g. `-ANONnameM`, `-UNCLEARWORD`) are replaced with a neutral `[UNK]` token, because the anonymization codes encode the gender of the person being named and would leak information unrelated to the speaker's own language. The ten most prolific speakers produce 19.3% of all utterances, which is why speaker-level methods matter here.

Category	Items
Filler	uh, um, erm, er
Response	yeah, mm
Interactional	oh, well, so, like, right, okay, anyway, you know, I mean, mind you, you see
Hedge	sort of, kind of, maybe, perhaps, probably, possibly, I guess, I think, fairly, roughly, pretty much, in a way, more or less, to some extent, it seems, as far as I can tell
Booster	really, definitely, absolutely, totally, certainly, clearly, utterly, indeed, undoubtedly, completely, extremely, entirely, surely, literally, for sure, of course, in fact, without a doubt, to be honest, as a matter of fact

Table 1: Discourse-marker inventory (53 items).

3.2 Discourse-marker inventory

Table 1 lists the 53 items we analyze, grouped by function. Response tokens (*yeah*, *mm*) are not discourse markers in the narrowest sense, but they are frequent, pragmatically similar and often discussed alongside DMs, so we keep them as a separate category. We exclude items that are not DMs: sequencers (*first*, *second*), modal verbs (*might*), frequency and degree adverbs (*always*, *never*, *quite*) and the answer particles *yes* and *no*.

DMs are matched as whole token sequences, longest match first, so *so* never matches inside *also* and *you know* is counted once rather than as *you* plus *know*. Several forms are also ordinary content words, so we use their CLAWS tags to exclude some non-DM uses: *like* tagged as a verb (`VV*`), *well* and *right* tagged as adjectives or nouns, and *sort of* and *kind of* whose first token is tagged as a noun (*what sort of car*). These filters remove 19,674 verbal uses of *like*, 6,844 nominal *sort of* and 7,244 nominal *kind of*. Other ambiguous uses remain in the counts; Section 7 quantifies them. Of all utterances, 56.9% contain at least one inventory item.

4 Methods

4.1 Frequency analysis

For each DM we compute its rate per 10,000 tokens for women and for men, and the $\log_2$ ratio of the two rates (positive means more frequent among women). Significance is tested two ways. The corpus-level test is log-likelihood (G^2) on pooled counts (Rayson and Garside, 2000). The speaker-level test follows Brezina and Meyerhoff (2014): we compute each speaker's rate per 1,000 tokens, using the 598 speakers with at least 1,000 tokens, and compare women and men with a two-sided Mann–Whitney U test, reporting the rank-biserial correlation r_{rb} as the effect size ($r_{rb} > 0$ means higher among women). Both sets of p-values are corrected across the inventory with the Benjamini–Hochberg procedure (Benjamini and Hochberg, 1995) at $q < 0.05$. We treat the speaker-level test as the primary result.

4.2 Speaker-level classification

Each of the 598 speakers is represented by a vector of their 53 DM rates. We train an L2-regularized logistic regression on standardized rates and report balanced accuracy from five repetitions of stratified five-fold cross-validation. To test whether this beats chance, we compare it with 200 runs on randomly permuted labels.

4.3 Utterance-level classification

Splits. The task is to predict a speaker's gender from a single utterance containing at least one DM. Speakers, not utterances, are assigned to training (70%), validation (10%) and test (20%) sets, separately within each gender, so that no speaker appears in more than one set. Each set is then balanced by randomly downsampling the larger gender. Training uses 100,000 utterances from 464 speakers, validation 10,000 from 66, and test 98,216 from 135. Every model is evaluated on exactly the same test utterances.

Input conditions. Each utterance is presented in three forms: *full text*, *DMs removed* (every matched DM deleted) and *DMs only* (just the sequence of matched DMs, e.g. *yeah like you know*). If DMs carry gender information, the DMs-only condition should beat chance; if a full-text model mostly relies on other words, removing DMs should cost little. Removing DMs leaves 29.7% of utterances empty, since many turns consist of a DM alone (e.g. *yeah*).

Models. The logistic-regression baseline uses unigram and bigram counts. BERT (`bert-base-uncased`) is fine-tuned with a linear classification head for up to 3 epochs (learning rate 2×10^{-5}, batch size 32, maximum length 64 tokens, linear warm-up over 6% of steps, weight decay 0.01). We evaluate on the validation set after each epoch and keep the checkpoint with the lowest validation loss. To keep the three BERT runs tractable on one GPU, both models are trained on the same random sample of 50,000 training utterances per gender.

Evaluation. We report accuracy, macro-F1, ROC AUC and per-class recall. Because utterances from one speaker are not independent, 95% confidence intervals come from a speaker-cluster bootstrap: we resample test speakers with replacement within each gender, 1,000 times. Differences between two models are computed on the same resampled speakers, and we call a difference reliable when its interval excludes zero. As a second view, we average each model's predicted probabilities over all of a test speaker's utterances and report balanced accuracy at the speaker level.

4.4 Masked-language-model probe

To measure the associations pre-trained BERT brings to the task, we use seven templates in the style of Kurita et al. (2019), such as *"{DM}, [MASK] said it was fine."* and *"{DM}, I'm just a [MASK]."*. For each template we compute

$$s = \log \frac{\sum_{w \in \mathcal{M}} P(w)}{\sum_{w \in \mathcal{F}} P(w)}$$

where $\mathcal{M}$ and $\mathcal{F}$ are small sets of male and female words that fit the slot (*he/she*, *man/woman*, *husband/wife* and so on). A DM's shift is s with the DM minus s for the same template without it, averaged over templates; a negative shift means the DM moves BERT toward female words. We correlate these shifts, with the sign flipped, with the corpus $\log_2$ ratios using Spearman's ρ.

5 Results

5.1 How different are women's and men's DM rates?

Pooled counts suggest that DM use is pervasively gendered: 37 of the 53 items differ significantly after correction. At the speaker level, only 10 do, 7 used more by women and 3 by men, and every one

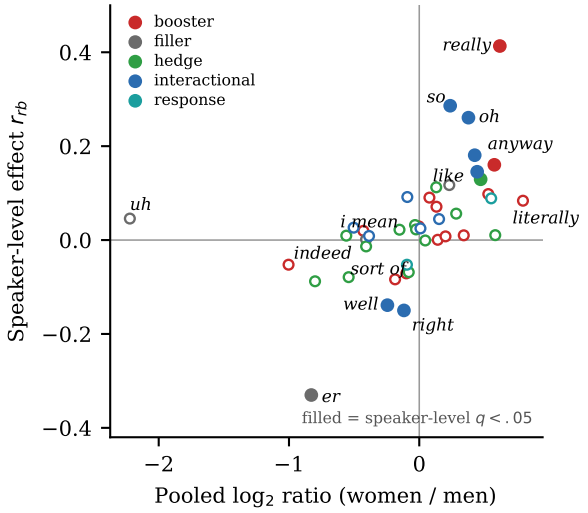


Figure 1: Pooled $\log_2$ ratio of each DM's rate (women/men) against its speaker-level effect size. DMs with at least 200 occurrences. Filled points differ significantly at the speaker level (Mann–Whitney, BH-corrected $q < 0.05$). Many DMs with sizeable pooled ratios, such as *uh*, show no speaker-level difference.

of these is also significant in the pooled test. The other 27 pooled "differences" are not supported once speakers are the unit of analysis (Figure 1). The filled pause *uh* is the starkest case: in pooled counts men use it 4.7 times as often as women, yet the typical man and the typical woman do not differ ($r_{rb} = +0.05$, $q = .36$). A minority of male speakers who use *uh* heavily produce the pooled result. *Sort of*, *I mean*, *um* and *of course* follow the same pattern, all pooled-significant with $|r_{rb}| \leq 0.03$. Appendix Table 4 gives both tests for every DM with at least 200 occurrences.

The reliable differences are modest. The largest is *really* ($r_{rb} = +0.41$; 49.0 vs. 31.9 per 10,000 tokens), followed by *er* ($r_{rb} = -0.33$; 35.1 vs. 62.4), *so* (+0.29) and *oh* (+0.26). An r_{rb} of 0.41 means that a randomly chosen woman uses *really* at a higher rate than a randomly chosen man about 70% of the time. *Like*, *anyway*, *clearly* and *kind of* lean slightly toward women, and *well* and *right* slightly toward men.

Table 2 aggregates by category. Women use more boosters ($r_{rb} = +0.35$) and more interactional markers (+0.25). Hedges show no difference (+0.05, $q = .43$), and neither do response tokens. Fillers lean toward men, but not significantly once corrected (−0.10, $q = .07$).

Category	Women	Men	$\log_2$	r_{rb}	q
Booster	65.8	47.3	+0.48	+0.35	<.001
Interactional	484.0	429.1	+0.17	+0.25	<.001
Hedge	65.3	63.0	+0.05	+0.05	.43
Response	304.0	292.9	+0.05	−0.01	.76
Filler	97.4	125.6	−0.37	−0.10	.07

Table 2: DM use by category: rate per 10,000 tokens, pooled $\log_2$ ratio (women/men), and speaker-level Mann–Whitney effect size and BH-corrected q.

5.2 Can a speaker's DM rates predict their gender?

Yes, moderately. From the 53 DM rates alone, logistic regression predicts a speaker's gender with a balanced accuracy of 0.726 (±0.039 across folds) on 598 speakers. None of 200 permuted-label runs did as well (permutation $p = 0.005$, the smallest value 200 permutations can give; null mean 0.499). The largest coefficients point to the same markers as the frequency analysis: *really*, *oh*, *so* and *clearly* toward women, and *er*, by a wide margin, toward men.

5.3 Utterance-level classification

Table 3 reports all models on the shared test set of unseen speakers. Every model is only a few points above chance. Full-text BERT is the most accurate at 0.576 [0.539, 0.609], but its advantage over logistic regression (0.557) is not reliable: +1.8 points [−0.7, +4.5].

The ablations show how little of this signal comes from DMs. With DMs alone, BERT reaches 0.541 and logistic regression 0.550, reliably but only slightly above chance (BERT: +4.0 points [+0.7, +7.3]). Removing the DMs does not reliably hurt either model: full-text BERT is +0.6 points more accurate than BERT without DMs [−3.9, +5.0], and logistic regression changes by +0.4 [−4.3, +5.0]. Whatever the full-text models use is available without the DMs. The most predictive features of the full-text logistic regression are content words, such as *granny*, *gorgeous* and *gosh* for women and *mate*, *movie* and *aye* for men, which reflect topics and regional or social vocabulary more than how talk is structured.

Training curves agree. For the full-text and DMs-removed models, validation loss is lowest after the first epoch and rises afterwards while training loss falls, a sign of memorization rather than generalizable signal (Appendix Figure 2). The DMs-only model's best validation loss, 0.687, is barely below

Model	Input	Accuracy [95% CI]	Macro-F1	AUC	Speaker acc.
Chance	–	0.501 [0.498, 0.504]	0.501	0.501	0.482
LR	full text	0.557 [0.532, 0.579]	0.557	0.589	0.774
LR	DMs removed	0.553 [0.508, 0.598]	0.543	0.571	0.824
LR	DMs only	0.550 [0.521, 0.572]	0.549	0.573	0.678
BERT	full text	0.576 [0.539, 0.609]	0.573	0.626	0.619
BERT	DMs removed	0.570 [0.503, 0.635]	0.548	0.611	0.624
BERT	DMs only	0.541 [0.508, 0.575]	0.536	0.571	0.688

Table 3: Gender prediction from single utterances, evaluated on 98,216 test utterances from 135 speakers not seen in training (balanced, so chance is 0.5). Accuracy intervals are 95% speaker-cluster bootstrap CIs. *Speaker acc.* is balanced accuracy after averaging each test speaker's predicted probabilities over all of their utterances.

the 0.693 of a model that always predicts 50/50.

Accuracy grows with utterance length. Full-text BERT is at chance on one- and two-word utterances (0.511) and reaches 0.640 on utterances of more than 20 words. Averaging over all of a speaker's utterances helps more: logistic regression identifies the gender of 0.774 of test speakers from full text, 0.824 with DMs removed and 0.678 from DMs alone. These speaker-level figures rest on only 135 test speakers and are correspondingly uncertain. BERT's speaker-level accuracy is lower (0.62–0.69). One likely reason is that its predictions are skewed toward one class; for example, BERT without DMs labels 79% of women's utterances female but only 35% of men's utterances male, and averaging carries that skew over to whole speakers.

The fine-tuned model also exaggerates real tendencies into near-rules. Full-text BERT labels 96% of test utterances containing *mm* as female, although 56% are; 85% of those with *erm* (true share 46%) and 83% of those with *really* (true share 63%). It labels *right* utterances female 78% of the time, although *right* leans male in the corpus and only 34% of those test utterances come from women.

5.4 What does pre-trained BERT associate with each DM?

Before any fine-tuning, BERT's gender associations for DMs line up with real usage. Across the 47 DMs with at least 50 corpus occurrences, DMs used relatively more by women also shift BERT's predictions toward female words ($\rho = 0.45$, $p = 0.001$). *Oh* and *really*, the two strongest female-leaning markers in the corpus, produce the largest shifts toward female words; *roughly*, *fairly*, *uh* and *yeah*, all male-leaning in pooled counts, shift BERT toward male words.

6 Discussion

DM use is gendered on average, not in the individual turn. Across a speaker's whole recorded speech, DM rates carry real information about gender: 0.726 balanced accuracy, driven by a handful of markers. In any single utterance, though, the signal almost disappears; models given only the DMs are right about 54–55% of the time. This is the pattern Cameron (2007) and Bamman et al. (2014) describe: average differences can be reliable while the overlap between individual women and men is large. A difference that a classifier can exploit across hundreds of turns is not a difference listeners could use to tell who is speaking.

Pooled significance overstates gender differences. The drop from 37 to 10 significant DMs is the clearest result of the study and a direct confirmation of Brezina and Meyerhoff (2014). Pooled tests in a corpus where the ten most prolific speakers produce a fifth of all utterances mostly measure those speakers. The *uh* case, a nearly fivefold pooled difference with no speaker-level difference, shows how a striking corpus statistic can describe a few individuals. Claims about gendered DM use that rest on pooled frequency tests should be treated with caution. The contrast between *er* (strongly male-leaning per speaker) and *uh* (no speaker-level difference) is also a reminder that two filled pauses can pattern differently, consistent with Tottie (2011), although transcription conventions for filled pauses may contribute.

Revisiting Lakoff. Lakoff (1973) predicted that women would use more hedges and more intensifiers. Fifty years later, in everyday British con-

versation, the intensifier half of the claim holds in weakened form: women use more boosters, especially *really* and *so*. The hedging half does not; women and men hedge at the same rate. Neither result tests Lakoff's broader point that perceptions of women's speech are shaped by who is speaking rather than only by what is said, which frequency data cannot address.

What the classifiers learn. Utterance-level models relied mainly on content words, and fine-tuned BERT turned modest population tendencies into near-categorical rules (*mm* $\rightarrow$ female) and sometimes into wrong ones (*right*). Because pre-trained BERT already associates female-leaning DMs with female words, fine-tuning starts from, and may amplify, an association that is real in aggregate but unreliable for individuals. This is the risk of using such a model to make judgments about specific speakers.

7 Limitations

Matching precision. Several DMs are polysemous, and our filters remove only some non-DM uses. The largest remaining source of noise is *like*: of the 137,784 tokens counted, 60,714 (44%) are tagged as prepositions (*look like*, *something like that*), and many others, tagged as adverbs, are the quotative *be like* (*and I was like, oh okay*), a distinct and itself gendered sociolinguistic variable. Counts for *so* include 16,491 degree-adverb uses (*so tired*; 16%), and counts for *well* include 9,184 tokens of *as well* (13%). Rates are per token including punctuation, which slightly lowers all rates equally. Utterances with low-confidence speaker attribution were not removed. A rerun with stricter context rules, including a separate quotative category, would sharpen the estimates for *like*, *so* and *well*; we would not expect it to change the overall conclusions, which rest on many DMs.

Scope. Gender here is a binary, self-reported metadata field; the corpus offers no basis for analyzing non-binary speakers, and our results say nothing about gender identity beyond that field. The corpus is informal British English recorded in 2012–2016, and results may not transfer to other varieties or settings. Gender is confounded with other speaker attributes (age, region, social class) and with conversational context, such as who a speaker is talking to; we do not control for these.

Models. BERT was trained on a subsample of 100,000 utterances with a single random seed, and speaker-level accuracies rest on 135 test speakers. The probe uses seven templates and small word lists.

Ethics Statement

Predicting a person's gender from their speech can be used to profile people without consent and reinforces a binary view of gender. We build classifiers only to measure how much gender information DMs carry, and we do not release trained models. The corpus was collected with participants' consent and is anonymized; we additionally neutralize anonymization codes that encode gender.

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A Additional Results

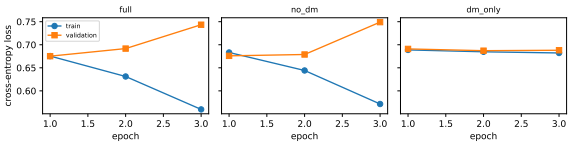


Figure 2: BERT training and validation loss by epoch for each input condition. The checkpoint with the lowest validation loss is used for testing.

DM	Category	Women per 10k words	Men per 10k words	$\log_2$ ratio	r_{rb}
literally	booster	2.2	1.2	$+0.80^{***}$	$+0.08$
really	booster	49.0	31.9	$+0.62^{***}$	$+0.41^{***}$
in a way	hedge	0.6	0.4	$+0.58^{***}$	$+0.01$
clearly	booster	0.5	0.3	$+0.58^{***}$	$+0.16^{***}$
mm	response	82.1	56.0	$+0.55^{***}$	$+0.09$
completely	booster	1.7	1.2	$+0.53^{***}$	$+0.10$
kind of	hedge	4.3	3.1	$+0.47^{***}$	$+0.13^{*}$
like	interactional	134.2	98.6	$+0.44^{***}$	$+0.15^{*}$
anyway	interactional	6.4	4.8	$+0.43^{***}$	$+0.18^{**}$
oh	interactional	92.7	71.3	$+0.38^{***}$	$+0.26^{***}$
absolutely	booster	2.7	2.2	$+0.34^{***}$	$+0.01$
maybe	hedge	8.0	6.6	$+0.28^{***}$	$+0.06$
so	interactional	96.7	82.0	$+0.24^{***}$	$+0.29^{***}$
erm	filler	52.5	44.7	$+0.23^{***}$	$+0.12$
to be honest	booster	1.0	0.8	$+0.20^{*}$	$+0.01$
okay	interactional	20.3	18.2	$+0.15^{***}$	$+0.04$
entirely	booster	0.2	0.2	$+0.14$	$+0.00$
definitely	booster	3.0	2.7	$+0.13^{*}$	$+0.07$
i think	hedge	32.0	29.2	$+0.13^{***}$	$+0.11$
surely	booster	0.5	0.5	$+0.08$	$+0.09$
perhaps	hedge	0.9	0.8	$+0.05$	-0.00
mind you	interactional	0.5	0.5	$+0.01$	$+0.02$
totally	booster	0.8	0.8	-0.00	$+0.03$
i guess	hedge	1.8	1.8	-0.02	$+0.02$
it seems	hedge	0.4	0.5	-0.03	$+0.03$
probably	hedge	9.9	10.5	-0.08^{**}	-0.07
you know	interactional	39.1	41.7	-0.09^{***}	$+0.09$
yeah	response	221.9	236.9	-0.09^{***}	-0.05
certainly	booster	0.7	0.7	-0.10	-0.07
right	interactional	20.4	22.1	-0.12^{***}	-0.15^{*}
possibly	hedge	0.7	0.8	-0.15	$+0.02$
in fact	booster	0.7	0.8	-0.19	-0.08
well	interactional	56.0	66.3	-0.24^{***}	-0.14^{*}
i mean	interactional	15.2	19.9	-0.39^{***}	$+0.01$
sort of	hedge	5.6	7.4	-0.41^{***}	-0.01
um	filler	8.1	10.7	-0.41^{***}	$+0.00$
of course	booster	2.4	3.2	-0.43^{***}	$+0.02$
you see	interactional	2.6	3.7	-0.50^{***}	$+0.03$
pretty much	hedge	0.6	0.8	-0.54^{***}	-0.08
more or less	hedge	0.2	0.2	-0.56^{**}	$+0.01$
fairly	hedge	0.4	0.7	-0.80^{***}	-0.09
er	filler	35.1	62.4	-0.83^{***}	-0.33^{***}
indeed	booster	0.2	0.4	-1.00^{***}	-0.05
uh	filler	1.7	7.7	-2.22^{***}	$+0.05$

Table 4: DMs with at least 200 occurrences, sorted by pooled $\log_2$ ratio (women/men). Stars on the log ratio mark the pooled log-likelihood test; stars on r_{rb} mark the speaker-level Mann–Whitney test (BH-corrected; $^{*}q < .05$, $^{**}q < .01$, $^{***}q < .001$).